Fix the universe and properties

Let finite U have size N, with properties $A_1, \ldots, A_n \subseteq U$. An intersection over J means satisfying at least all properties in J; properties outside J may also hold. Define $S_0 = N$ and

$$S_k = \sum_{\substack{J \subseteq [n] \\ |J| = k}} \left| \bigcap_{i \in J} A_i \right|.$$

This sums intersection sizes, rather than measuring their union. An object satisfying c properties appears $\binom{c}{k}$ times in S_k. This multiplicity explains alternating signs, exact-hit formulas, and truncation errors.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

General proof by indicators

Let I_i(x) indicate membership in A_i. The indicator of no hit is $\prod_i (1 - I_i(x))$. Expand and sum over U:

$$\left| U \setminus \bigcup_i A_i \right| = \sum_{J \subseteq [n]} (-1)^{|J|} \left| \bigcap_{i \in J} A_i \right| = \sum_{k=0}^{n} (-1)^k S_k.$$

The empty intersection is U, supplying N. Finite sums commute and each indicator product counts its intersection. Neither independence nor equal intersection sizes is required. Subtracting from N gives union size $S_1 - S_2 + S_3 - \ldots$. At n=0, the union is empty and the no-hit count N, consistent with empty product one.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Why each union member remains once

For c≥1 hits, net multiplicity is

$$\sum_{k=1}^{c}(-1)^{k+1}\binom{c}{k} = 1-(1-1)^c = 1.$$

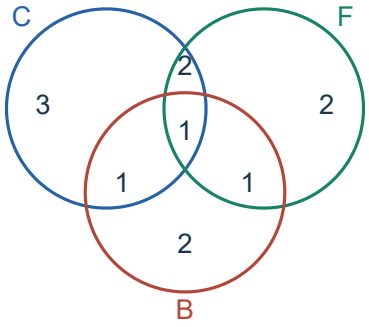
Objects with c=0 enter no positive-order intersection and contribute zero. Four hits contribute 4−6+4−1=1; stopping after pair intersections leaves −2. Multiple overlaps require the full accounting. Individual set sizes alone do not determine intersections; inclusion–exclusion organizes known data, rather than supplying missing data.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Every region for three defects

Among 20 nodes, defects C,F,B have sizes 7,6,5; pair intersections 3,2,2; triple intersection 1. Pair intersections include triple members.

Numbers label disjoint regions, not entire intersections



No hits: 8

N=20 ; |C|=7 · |F|=6 · |B|=5

|C∩F|=3 · |C∩B|=|F∩B|=2

$S_1=18$ · $S_2=7$ · $S_3=1$

At least one: 18−7+1=12

No hits: 20−12=8

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Counting each disjoint region

Exactly two defects give 2,1,1. Only C gives 7−3−2+1=3; only F gives 2 and only B gives 2. Interior regions total 3+2+2+2+1+1+1=12, leaving 8 compliant nodes. Equivalently $S_1=18, S_2=7, S_3=1$ and 20−18+7−1=8. A negative disjoint region would expose inconsistent input intersections, even if an alternating arithmetic expression could still be evaluated.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Truncated alternating binomial sums

For integers $c \geq 1, r \geq 0$, with out-of-range binomials zero,

$$\sum_{k=0}^{r} (-1)^k \binom{c}{k} = (-1)^r \binom{c-1}{r}.$$

Substitute Pascal's identity $\binom{c}{k} = \binom{c-1}{k} + \binom{c-1}{k-1}$. Shift the second sum by one; internal terms cancel, leaving the final term. For $r \geq c$ both sides vanish.

Thus an object with $c \geq 1$ hits contributes $1 - (-1)^r \binom{c-1}{r}$ to the union sum through order r. It is at least one for odd r and at most one for even r; outside objects always contribute zero. This proves the direction rather than guessing from signs.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Bonferroni bounds

For $T_r = \sum_{k=1}^{r} (-1)^{k+1} S_k$, 1≤r≤n, summing the pointwise inequalities gives

$$r \text{ odd}: \quad |\cup A_i| \le T_r, \qquad r \text{ even}: \quad |\cup A_i| \ge T_r.$$

No-hit bounds reverse on subtracting from N. These bounds require consistent intersection data through order r and do not require independence. In the defect example, T_1=18 is an upper bound and T_2=11 a lower bound. The union lies in [11,18], compliance in [2,9]. Order three gives exact values 12,8. Truncation is already exact when every higher-order intersection is empty.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Probability and weights use linearity

Taking expectations of the same finite indicator identity yields

$$P(\cup_i A_i) = \sum_{\varnothing \ne J \subseteq [n]} (-1)^{|J|+1} P(\cap_{i \in J} A_i).$$

Intersection probabilities need not factor. Mutual independence permits that substitution; pairwise independence alone does not give all higher-order products.

Replace cardinality by $\sum_{x \in A} w(x)$. For nonnegative weights the identity and Bonferroni directions survive by multiplying pointwise inequalities and summing. Signed weights retain the algebraic identity but need not retain inequality directions.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Divisibility intersections use lcm

Let M be a nonnegative integer and a, b, d be positive integers. In 1,…,M, let A_d contain multiples of d. Then $A_a \cap A_b = A_{\operatorname{lcm}(a,b)}$, of size $\lfloor M / \operatorname{lcm}(a,b) \rfloor$. Only coprime divisors permit replacing lcm by product.

For M=40 avoiding 2,3,5, single counts are 20,13,8; pair counts 6,4,2; triple 1. Thus

$$40 - (20 + 13 + 8) + (6 + 4 + 2) - 1 = 10.$$

The numbers are 1,7,11,13,17,19,23,29,31,37. Multiples of both 4 and 6 are multiples of 12, not 24. Alternating signs cannot correct a wrongly modeled intersection.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Euler's totient product from inclusion–exclusion

For a positive integer m, the totient φ(m) counts integers in 1,…,m coprime to m. Let distinct prime divisors be p_1,…,p_s. Noncoprimality is equivalent to divisibility by at least one of them, since any common divisor above one has a prime divisor. Every subset product d divides m, so its intersection has exactly m/d members, without floor error. Expanding gives

$$\varphi(m) = m \prod_{p \mid m} (1 - 1/p).$$

For m=30, 30(1/2)(2/3)(4/5)=8, with 1,7,11,13,17,19,23,29. At m=1 the empty product gives φ(1)=1. For a general upper endpoint M, retain floors; this product cannot be applied unconditionally.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Bounded nonnegative integer solutions

For m≥1, N,b_i nonnegative integers, count $\sum_i x_i = N, 0 \le x_i \le b_i$. Without upper bounds there are $\binom{N+m-1}{m-1}$ solutions. If every i∈J violates its bound, subtract b_i+1 from x_i. This is a bijection with nonnegative solutions of the reduced total, giving

$$\# = \sum_{J \subseteq [m]} (-1)^{|J|} \binom{N - \sum_{i \in J}(b_i + 1) + m - 1}{m - 1}.$$

A negative remaining total means zero solutions; do not interpret it through generalized negative-upper binomials. For $x_1+x_2+x_3=6$ with bounds 2,3,4, the unrestricted count is 28. Single violations leave totals 3,2,1, giving 10,6,3. Every double violation needs more than six. Thus 28−10−6−3=9.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Surjections exclude missed outputs

For nonnegative integers m and n, there are n^m functions from m points to n points. Let A_i mean output i is missed. Missing a specified k outputs leaves (n−k)^m functions, so

$$\operatorname{Surj}(m, n) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n - k)^m = n! S(m, n).$$

The partition expression counts nonempty fibers, and the alternating expression follows from the general proof. For m<n the count is zero; m=n gives n!. For m=0 the unique empty function is surjective only when n=0: the count is 0 for n>0 and 1 for m=n=0. The counting convention $0^0 = 1$ includes the empty function in the formula. Five labeled inputs onto three named outputs give 243−3 • 32+3 • 1−0=150. Exactly two outputs instead requires choosing them first: $\binom{3}{2}(2^5 - 2) = 90$. Deleting a term of the onto-three formula does not change its meaning correctly.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Circular arrangements require valid blocks

For m≥2 disjoint specified pairs among 2m people, identify rotations but distinguish reflections. If k specified pairs are adjacent, contract each pair into a labeled block. There remain 2m−k distinct objects around the circle and two orders inside each block, giving $2^k(2m-k-1)!$. Hence no specified pair adjacent gives

$$\sum_{k=0}^{m}(-1)^k\binom{m}{k}2^k(2m-k-1)!.$$

At m=4 the terms are 5040−5760+2880−768+96=1488. The condition 2m≥4 excludes the two-person circular degeneracy in which two orientations represent one rotation class. If bad pairs share people, independent two-person contractions are invalid.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

No isolated vertices need not mean connected

There are $2^{\binom{n}{2}}$ labeled undirected simple graphs. If specified k vertices must be isolated, only edges among the other n−k vertices remain free, giving $2^{\binom{n-k}{2}}$. Thus the count without isolated vertices is

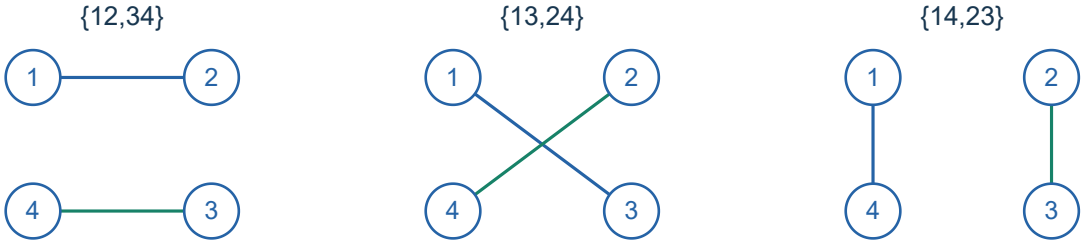
$$\sum_{k=0}^{n}(-1)^k\binom{n}{k}2^{\binom{n-k}{2}}.$$

For n=4 this is 64−32+12−4+1=41. Three are disconnected perfect matchings.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

The four-vertex example

Four labeled vertices: exactly three disconnected graphs without isolates



All degrees are 1; crossings are not vertices. Connected graphs: 41−3=38.

Every disconnected no-isolate graph on four vertices has two components of size two, each with its unique edge. These three exhaust the cases, so connected graphs number 41−3=38. Inclusion–exclusion answers the stated bad properties, not a substituted graph property.

Intersection sums and exact-hit counts

Let E_t count objects satisfying exactly t properties. An object with c hits contributes $\binom{c}{k}$ to S_k, so $S_k = \sum_{c=k}^{n} \binom{c}{k} E_c$. Define the finite polynomial $G(z) = \sum_{x \in U} z^{c(x)} = \sum_t E_t z^t$. Expand $z^{c(x)} = [1 + (z-1)]^{c(x)}$ and interchange finite sums:

$$G(z) = \sum_{k=0}^{n} S_k (z-1)^k.$$

This polynomial identity has no convergence condition. At z=1 it gives N; at z=0 the no-hit count. Coefficient comparison gives inversion.

Exact-hit inversion with a pointwise proof

The coefficient of z^t gives

$$E_t = \sum_{k=t}^{n} (-1)^{k-t} \binom{k}{t} S_k.$$

An object with c<t contributes zero, and one with c=t contributes one. For c>t,

$$\sum_{k=t}^{c} (-1)^{k-t} \binom{k}{t} \binom{c}{k} = \binom{c}{t} \sum_{j=0}^{c-t} (-1)^{j} \binom{c-t}{j} = 0.$$

Here $\binom{k}{t}\binom{c}{k} = \binom{c}{t}\binom{c-t}{k-t}$: both choose t distinguished elements and k−t extra ones. Thus precisely the target objects remain. The cases t=0 and t=n give inclusion–exclusion and E_n=S_n respectively.

At-least-hit coefficients differ

For 1≤t≤n, let $H_t = \sum_{j=t}^{n} E_j$. Substituting inversion and swapping sums gives the coefficient of S_k as

$$\sum_{j=t}^{k} (-1)^{k-j} \binom{k}{j} = (-1)^{k-t} \binom{k-1}{t-1},$$

by the truncated binomial identity with index k−j. Therefore

$$H_t = \sum_{k=t}^{n} (-1)^{k-t} \binom{k-1}{t-1} S_k.$$

At t=1 this is the union formula. At least zero properties means N directly; do not substitute t=0 into a lower index −1. Exact and at-least formulas cannot be interchanged by merely dropping terms.

Recovering the full three-property distribution

For N=20,S_1=18,S_2=7,S_3=1,

$$E_3 = 1, \quad E_2 = 7 - 3 = 4, \quad E_1 = 18 - 2 \cdot 7 + 3 \cdot 1 = 7, \quad E_0 = 8.$$

The total 8+7+4+1=20. Reconstruct S_1=7+2 • 4+3 • 1=18 and S_2=4+3 • 1=7. At least two gives H_2=S_2−2S_3=5=E_2+E_3. No hits is the union complement, size 8; the triple-intersection complement has size 19 and answers another question.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Derangements and their formula

A permutation π:[n]→[n] is a bijection. It is a derangement if every π(i)≠i; write its count D_n. Let A_i mean i is fixed. Specifying k fixed points leaves (n−k)! permutations, so

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

The empty permutation gives D_0=1, while D_1=0. No fixed point differs from merely not being the identity. Swapping 1,2 while fixing the rest satisfies only the latter when fixed points remain.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Two reversible classes prove the recurrence

For $n \geq 2$ fix $\pi(1)=j \neq 1$, with $n-1$ choices. If $\pi(j)=1$, delete the two-cycle $(1\ j)$, leaving an $(n-2)$-point derangement.

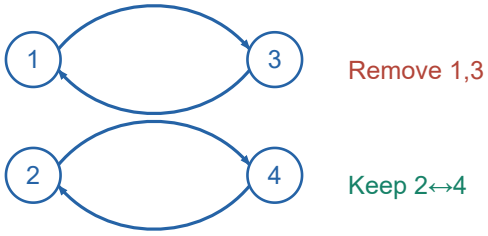
Otherwise let k be the unique predecessor of 1, so $k \neq j$. Contract $k \to 1 \to j$ to $k \to j$ and delete 1. No new fixed point occurs: the only possibility would require $k=j$, already excluded. Conversely, in any derangement of the remaining $n-1$ labels, insert 1 just before j at its unique predecessor. This uniquely restores the second class.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

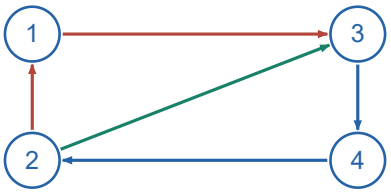
The two derangement cases

Fix $\pi(1)=3$: two-cycle deletion or longer-cycle contraction

Two-cycle: remove (1 3)



Longer cycle: contract 2→1→3 to 2→3



Remove red edges; add green. Inverse: insert 1 before 3.

The classes are disjoint and exhaustive, proving $D_n = (n-1)(D_{n-1} + D_{n-2})$. The chosen j remains the image of 1; do not silently relabel it as another point in the reduced problem.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Integer traces for five-point derangements

$D_0=1, D_1=0$ give $D_2=1, D_3=2, D_4=3(2+1)=9, D_5=4(9+2)=44, D_6=5(44+9)=265$.

k	$\binom{5}{k}(5-k)!$	Sign
0	120	+
1	120	−
2	60	+
3	20	−
4	5	+
5	1	−

Thus $120-120+60-20+5-1=44$, matching the recurrence. A permutation with three fixed points appears in several intersection sums; these rows are not disjoint permutation classes.

Distance from n!/e and nearest integers

The alternating remainder in $e^{-1}=\sum_{k\ge0}(-1)^k/k!$ gives, for $n\ge1$,

$$|D_n - n!/e| < 1/(n+1).$$

Even n lies above n!/e, odd n below. For $n\ge2$ the error is below one half; n=1 uses $1/e<1/2$ directly. Thus

$$D_n = \lfloor n!/e + 1/2 \rfloor \qquad (n \ge 1).$$

At n=0 use $D_0=1$ separately. Uniform-permutation probability $D_n/n!$ tends to e^{-1} rather than equaling it for every finite n. Numeric rounding also needs controlled computational error; the mathematical remainder alone is no floating-point guarantee.

Exactly fixed points and relative derangements

Choose t fixed points, then derange all others, giving $\binom{n}{t} D_{n-t}$. These disjoint exhaustive classes give $n! = \sum_t \binom{n}{t} D_{n-t}$. For n=5, counts at t=0,…,5 are 44,45,20,10,0,1, totaling 120. Exactly n−1 fixed points is impossible: bijectivity forces the last point fixed too.

For two assignment rounds with no person receiving the same item twice, fix any first-round bijection and relabel by it. The second is a relative derangement, giving n!D_n ordered pairs of rounds. It is not D_n^2 because the first round has no original-label restriction.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Rook polynomials on a specified board

For $B \subseteq [m] \times [n]$, r_k(B) counts k rooks in B using distinct rows and columns; r_0=1 and $R_B(x) = \sum_k r_k(B) x^k$. At most min(m,n) rooks fit.

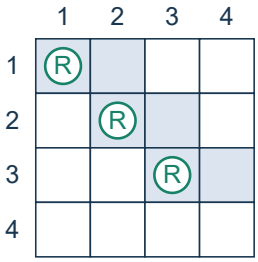
B specifies where rooks are counted. When B later represents forbidden assignment positions, rooks still lie in B to count bad-event intersections; B does not become allowed assignment cells. On a full m×n board, choose k rows, k columns, and a bijection: $r_k = \binom{m}{k}\binom{n}{k}k!$. The full 2×2 board gives $1+4x+2x^2$, not $(1+x)^4$, which would permit attacks.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Every coefficient of the six-cell ladder

In 4×4 take B={(1,1),(1,2),(2,2),(2,3),(3,3),(3,4)}.

Rooks on forbidden B: nonattacking cells give compatible bad events



B shaded; R: three rooks

$r_0=1$; $r_1=6$

$r_2=C(6,2)-3-2=10$

$r_3=4$; $r_4=0$

Three-rook column triples

(1,2,3) (1,2,4)

(1,3,4) (2,3,4)

At most one rook per row.

No fourth nonempty row, so four rooks are impossible.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Counting the ladder coefficients

$r_1=6$. Among $\binom{6}{2}=15$ cell pairs, three share rows and two share columns. Distinct cells cannot share both, so $r_2=10$. Three rooks require one in each of the first three rows. Their column triples are exactly (1,2,3),(1,2,4),(1,3,4),(2,3,4), giving $r_3=4$. Row four is empty, so $r_4=0$. Hence $R_B=1+6x+10x^2+4x^3$.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Delete a cell or select it

For e=(i,j)∈B, placements either omit e and lie in $B \setminus \{e\}$, or select e and avoid its entire row and column. Call the latter remaining board B_ij. Deleting or restoring e gives reversible disjoint cases:

$$r_k(B) = r_k(B \setminus \{e\}) + r_{k-1}(B_{ij}), \quad R_B = R_{B\setminus\{e\}} + xR_{B_{ij}}.$$

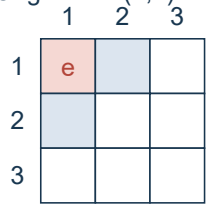
For B={(1,1),(1,2),(2,1)}, choose e=(1,1).

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

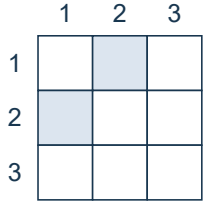
Deletion and selection example

Deletion/contraction: disjoint cases according to whether e is used

Original: e=(1,1)

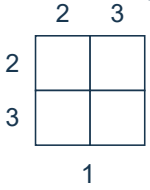


Omit e: delete one cell



1+2x+x²

Select e: delete row 1, column 1



R_B(x) = (1+2x+x²) + x·1 = 1+3x+x²

Omitting e leaves two nonattacking cells with polynomial $(1+x)^2$. Selecting e leaves an empty board, polynomial 1. Thus $R_B=1+2x+x^2+x=1+3x+x^2$. Each branch reduces cell count; empty-board base one ensures termination.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Board products need disjoint rows and columns

If $B=B_1 \cup B_2$ and both row sets and column sets are disjoint, a k-rook placement uniquely splits into i rooks in B_1 and k−i in B_2. Conversely two component placements cannot attack across boards. Hence

$$r_k(B) = \sum_{i=0}^{k} r_i(B_1) r_{k-i}(B_2), \qquad R_B = R_{B_1} R_{B_2}.$$

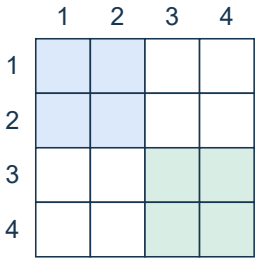
Cell disjointness alone fails. For $B_1=\{(1,1)\}, B_2=\{(1,2)\}$, each gives 1+x, but their product wrongly includes x^2; the union gives 1+2x. Splitting remains unique; combining individually legal placements fails. Row or column relabeling preserves attacks and bijects placements of each size, helping locate genuinely independent blocks.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Full convolution for two 2×2 blocks

Take only the upper-left and lower-right full 2×2 blocks of a 4×4 board. Their rows and columns separate.

Independent rook choices require disjoint rows and columns



Blue B_1: rows 1,2 × columns 1,2

Green B_2: rows 3,4 × columns 3,4

$R_1=R_2=1+4x+2x^2$

$R_B=1+8x+20x^2+16x^3+4x^4$

$r_2=2\cdot1+4\cdot4+1\cdot2=20$

Two rooks: both in B_1, one per block, or both in B_2.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Convolution of the block coefficients

$$R_B = (1 + 4x + 2x^2)^2 = 1 + 8x + 20x^2 + 16x^3 + 4x^4.$$

The x^2 coefficient is 2+4 • 4+2=20, for both rooks in one block or one in each. The x^3 coefficient is 4 • 2+2 • 4=16. Polynomial products count independent choices and require cross terms, not just matching-degree products.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Why forbidden-position intersections use rooks

A permutation chooses one cell in each row and column. For each forbidden (i,j)∈B, let its bad event be π(i)=j. If specified k cells share a row or column, their intersection is empty. Otherwise fixed nonattacking cells extend to (n−k)! bijections of remaining rows and columns. There are r_k(B) such cell sets, so

$$N_{\text{avoid}} = \sum_{k=0}^{n} (-1)^k r_k(B)(n-k)!.$$

This counts complete avoidance, not exactly k forbidden hits. A diagonal board gives R_B= (1+x)^n and the derangement formula. An empty board gives n!.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

All ladder-board avoiding permutations

The coefficients 1,6,10,4 at n=4 give

$$24 - 6 \cdot 6 + 10 \cdot 2 - 4 \cdot 1 = 4.$$

The row-ordered permutations are (3,1,2,4),(3,4,1,2),(3,4,2,1),(4,1,2,3). Row one permits only 3,4; row two forbids 2,3; row three forbids 3,4; each result uses every column once. Subtracting bad events independently repeats corrections to the same permutation. Rook coefficients organize jointly possible bad cells; factorials count their full extensions.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Rectangular injections retain a denominator

Assign m named jobs injectively to n named people, m≤n. Total assignments are (n)_m=n!/(n−m)!. Fixing k nonattacking forbidden cells leaves

$$(n-k)!/(n-m)!$$

injections of the other jobs. Therefore $N_{\text{avoid}} = \sum_{k=0}^{m} (-1)^k r_k(B)(n-k)!/(n-m)!$. Attacking cells still give empty intersections. At m>n there are no injections; return zero without negative factorials. At m=0 the unique empty assignment gives one. At m=n the denominator is 0!=1, recovering the square case.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Two jobs and four people

Take B={(1,1),(1,2),(2,2),(2,3)}. Here r_0=1,r_1=4; legal two-rook column pairs by row are (1,2), (1,3),(2,3), so r_2=3.

$$N = 4!/2! - 4 \cdot 3!/2! + 3 \cdot 2!/2! = 12 - 12 + 3 = 3.$$

Job one permits 3,4; job two permits 1,4. Remove the repeated-person choice (4,4), leaving (3,1),(3,4),(4,1). Omitting 2! counts each injection twice by unnecessarily ordering the two unused people.

Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Rook coefficients recover exact-hit distributions

Let h_t count permutations using exactly t cells of B. Since $S_k = r_k(B)(n-k)!$, inversion gives

$$h_t = \sum_{k=t}^{n} (-1)^{k-t} \binom{k}{t} r_k(B)(n-k)!.$$

The ladder has $S_0,\ldots,S_4$=24,36,20,4,0. Thus h_0=4,h_1=36−40+12=8,h_2=20−12=8,h_3=4,h_4=0, totaling 24. Also $S_1=h_1+2h_2+3h_3$=36. The forbidden-cell count r_1=6 differs from the exactly-one-hit permutation count 8; these quantities have different units.

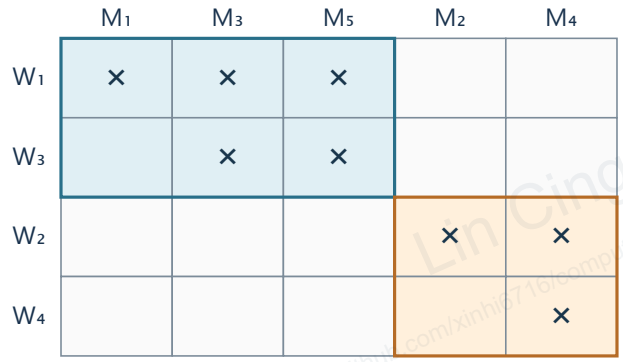
Lin Cing
https://github.com/xinhi6716/computer-science-self-study

Restricted matching: split the forbidden board into two blocks

Match each of four people $W_1, \dots, W_4$ to a different partner among $M_1, \dots, M_5$. Crosses mark forbidden matches.

Forbidden board: reorder rows and columns into independent blocks

× marks a forbidden match; labels retain their original indices



Block A: 2 rows × 3 columns

$R_a(x) = 1 + 5x + 4x^2$

Block B: 2 rows × 2 columns

$R_\beta(x) = 1 + 3x + x^2$

Restricted matching: coefficients and inclusion–exclusion

The forbidden columns for rows 1–4 are respectively {1,3,5}, {2,4}, {3,5}, {4}. Reordering changes only the display: the blocks share neither rows nor columns, so their rooks cannot attack each other.

Block A has five forbidden cells and four nonattacking two-rook placements; block B has three forbidden cells and one such pair. The product records all ways to choose rooks from both blocks:

$$R(x) = (1 + 5x + 4x^2)(1 + 3x + x^2) = 1 + 8x + 20x^2 + 17x^3 + 4x^4.$$

After fixing k nonattacking forbidden cells, the remaining $4 - k$ rows inject into $5 - k$ columns in $P(5 - k, 4 - k)$ ways. Thus the number of legal assignments is

$$P(5,4) - 8P(4,3) + 20P(3,2) - 17P(2,1) + 4P(1,0)$$
$$= 120 - 192 + 120 - 34 + 4 = \boxed{18}.$$

Here $P(a,b) = a!/(a-b)!$, and $P(1,0) = 1$.

Row-by-row rook dynamic programming

Let $d_i(S)$ count placements in the first i rows using exactly column set S. Initially $d_0(\varnothing) = 1$, others zero. For each row, omit a rook or select one available column j with $(i, j) \in B$:

```
new[...] = 0
for each subset S:
    new[S] += old[S]
    for each j with (i,j) in B and j not in S:
        new[S union {j}] += old[S]
old = new
```

Read only the preceding row's old table; in-place updates could place two rooks in one row.

Why the rook dynamic program works

Every legal placement uniquely determines whether this row has a rook, its column, and the previous placement. This proves completeness without duplication. Finally $r_k = \sum_{|S|=k} d_m(S)$.

There are at most $mn2^n$ additions and $O(2^n)$ counts in two layers. This is arithmetic-operation cost; large-integer bit cost is separate. Transpose to reduce the column dimension when helpful. This is not a polynomial-time claim for large boards.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 8.

Original educational synthesis by Lin Cing, licensed under CC BY 4.0.

https://github.com/xinhi6716/computer-science-self-study